

Analysis of Clinical, Biological and Prognostic Factors in Newly Diagnosed Chronic Lymphocytic Leukemia Patients

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Background

Chronic lymphocytic leukemia (CLL) has a well-described heterogeneous prognosis guided by clinical stage and biological markers. Outcomes vary widely between patients, and disease behaviour in North African / African populations remains comparatively under-documented relative to Western cohorts.

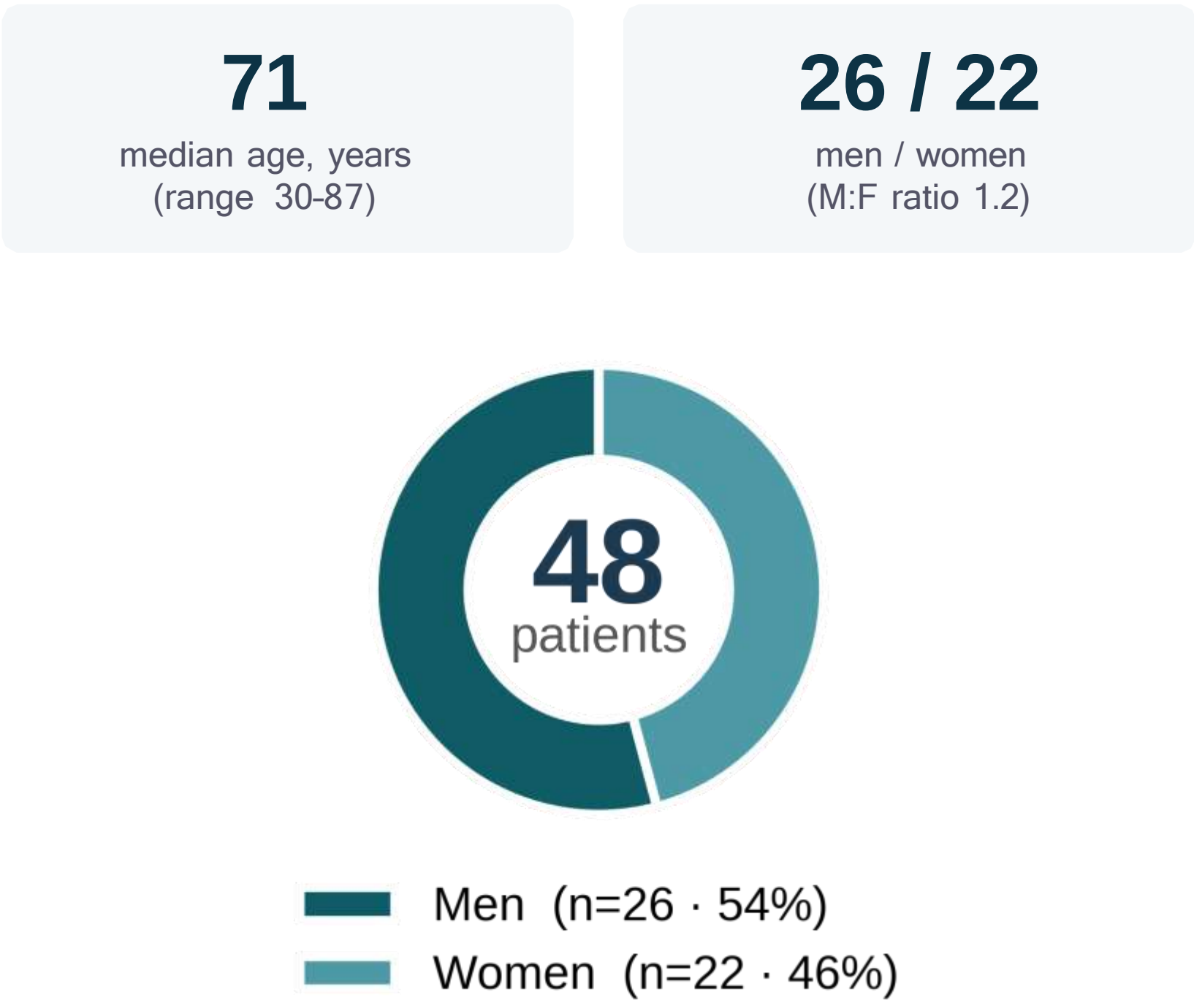
Objective

To describe the clinical presentation, biological profile, and distribution of documented prognostic factors in newly diagnosed Algerian patients with CLL.

Methods

48 adults newly diagnosed with CLL at the Hemobiology and Blood Transfusion Center of the Annaba University Hospital were analysed.

Study Population



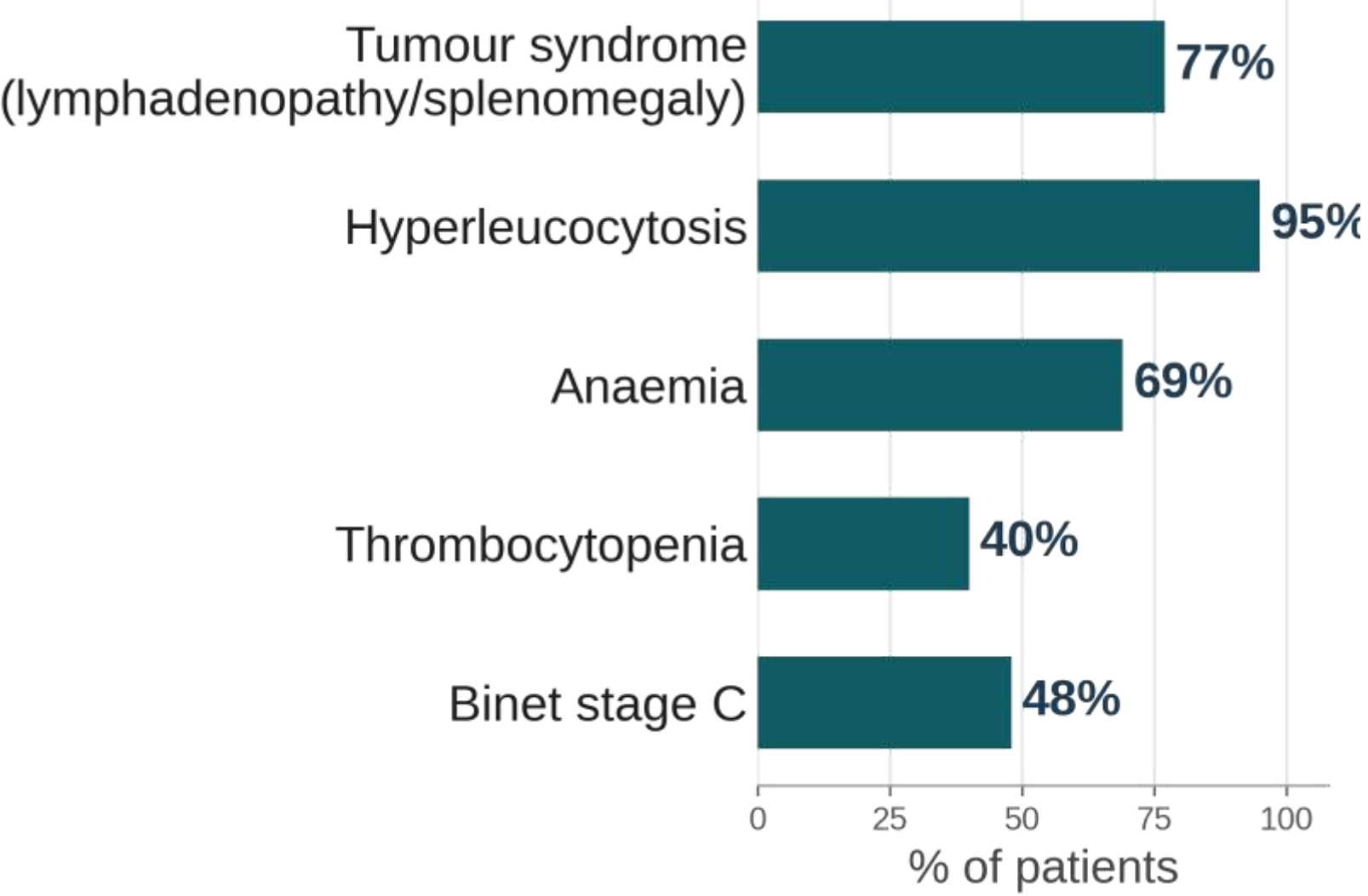
48

patients

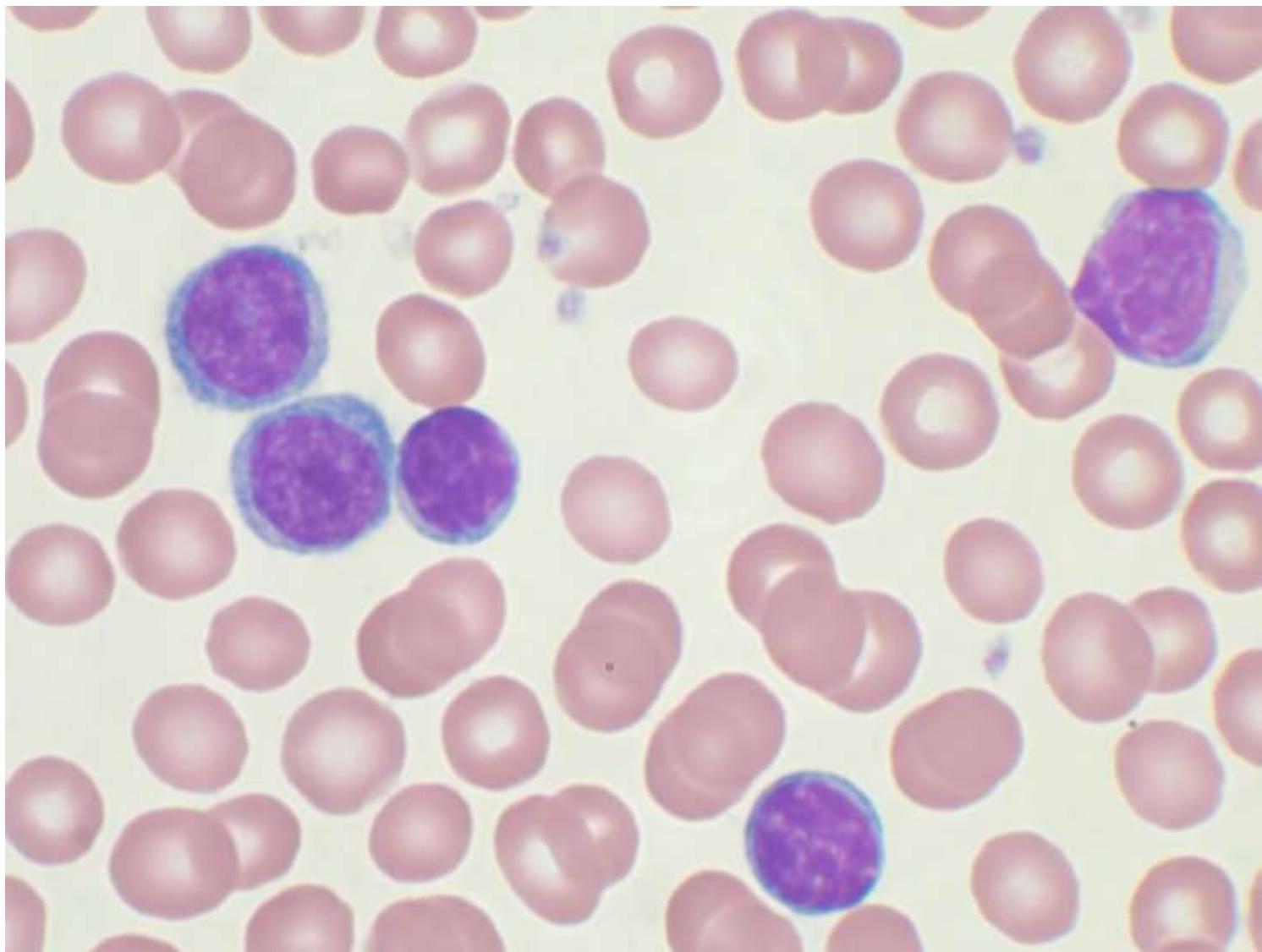
Men (n=26 · 54%)

Women (n=22 · 46%)

Clinical & Haematological Presentation



Peripheral Blood Smear



Lymphocytes in CLL

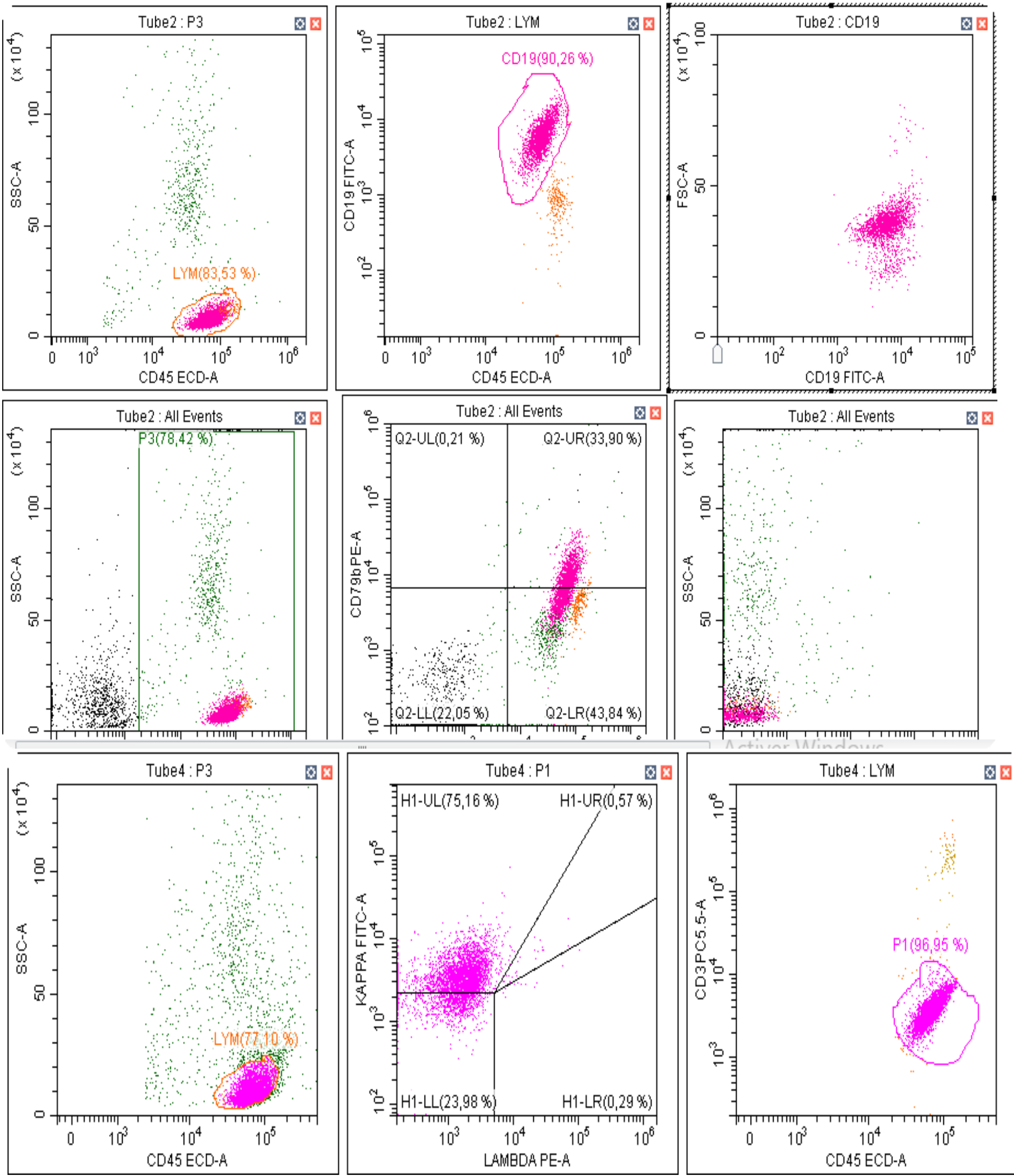
Diagnostic Score (Matutes)

Immunophenotypic scoring used to support the diagnosis of CLL

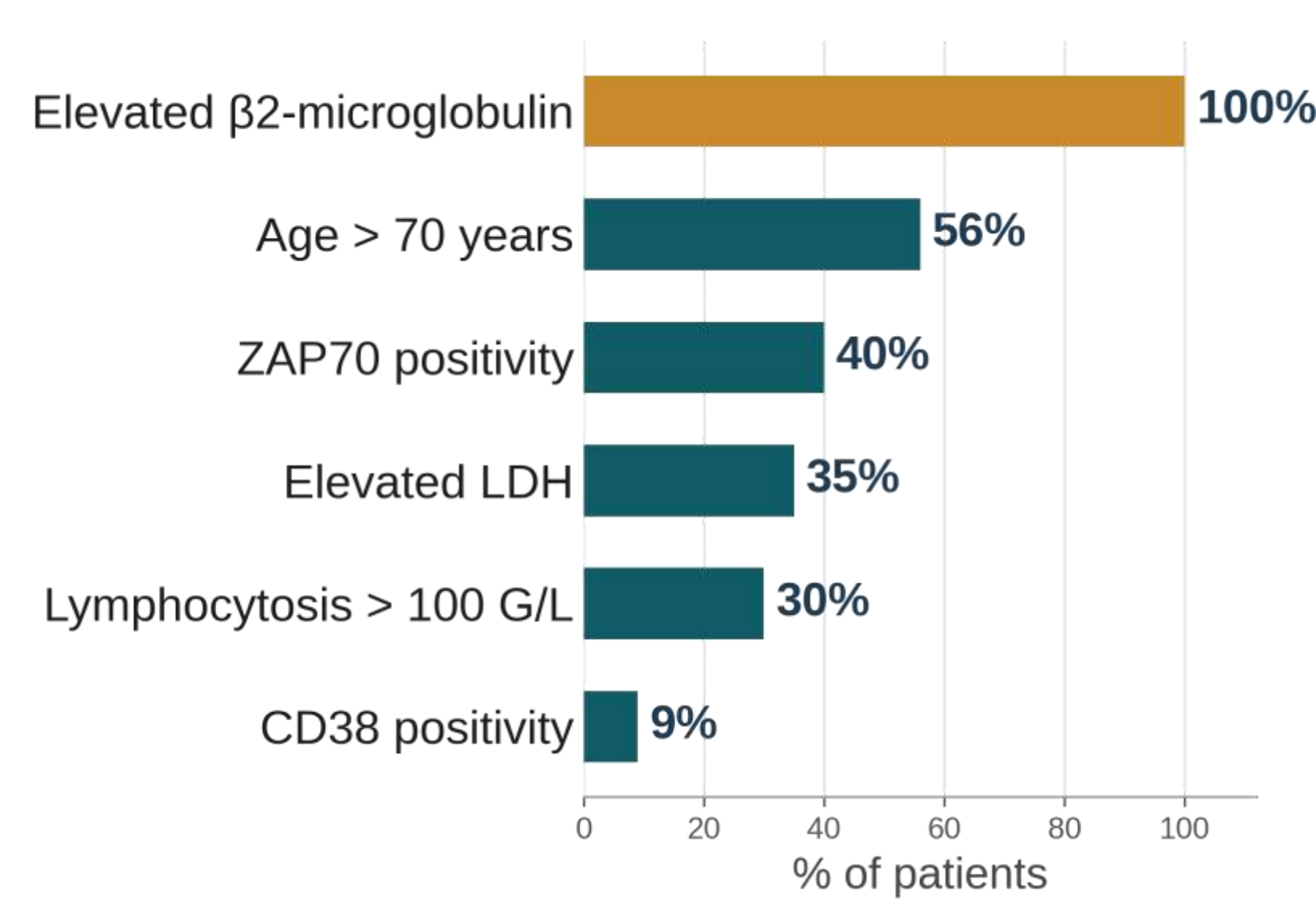
Marker	1 point	0 point
CD5	+	-
CD23	+	-
CD79b	Weak expression	Strong expression
FMC7	-	+
Surface Ig	Weak expression	Strong expression

A score of 4–5 is highly consistent with classical CLL.

Immunophenotyping (flow cytometry)



Distribution of Prognostic Factors



Prognostic factor	Frequency
Elevated β 2-microglobulin	100%
Age > 70 years	56%
ZAP70 positivity	40%
Elevated LDH	35%
Lymphocytosis > 100 G/L	30%
CD38 positivity	9%

Discussion

Our patients are characterized by a distinct profile: a high rate of tumour burden and marked cytopenias at diagnosis. Unlike Western populations but similar to other African cohorts, our patients present with **advanced and symptomatic disease**, a pattern that likely reflects delayed diagnosis in the region.

Nearly half of patients were already at Binet stage C – the most advanced clinical stage – at the time of first diagnosis.

Conclusion

Prognostic factors for CLL are designed to predict disease progression. Our findings emphasize the need to implement **early detection strategies** in the region to reduce diagnostic delay, improve outcomes, and preserve patients' quality of life.

Acknowledgements

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